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D025713(025)

B. Tech. (Seventh Semester) Examination, Nov.-Dec. 2022

(Scheme : AIC)

(EEE : Branch)

HIGH VOLTAGE ENGINEERING

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *All questions are compulsory. Part (a) of each question is compulsory carrying 4 marks. Attempt any* ***two*** *parts from (b), (c) and (d) of each question carrying 8 marks each.*

Unit-I

1. (a) Write the various factors that influence breakdown in a gas. 4

(b) Explain the Streamer theory of breakdown in air at atmospheric pressure. What are the anode and the cathode streamers? 8

(c) What is Pashchen's law? How do you account for the minimum voltage for breakdown under a given '$p \times d$' condition? 8

(d) In an experiment in a certain gas it was found that the steady state current is $5 \cdot 5 \times 10^{-8}$ A at 8 kV at a distance of 0.4 cm between the plane electrodes. Keeping the field constant and reducing the distance to 0·1 cm results in a current of $5 \cdot 5 \times 10^{-9}$ A. Calculate Townsend's primary ionization coefficient α. If the breakdown occurred when the gap distance was increased to 0·9 cm, what is the value of γ? 8

Unit-II

2. (a) What is thermal breakdown? 4

(b) Write a short note on 'Breakdown in commercial liquids' : 8

(i) Suspended particle theory

(ii) Bubble theory

(iii) Thermal mechanism of liquid breakdown

(c) Explain the phenomenon 'treeing and tracking' in solid insulating materials under electrical stress. How does it lead to breakdown? 8

(d) Explain on experimental method for obtaining breakdown strength of transformer oil. 8

Unit-III

3. (a) Define wave front and wave tails times of an impulse wave. 4

(b) A ten stage cockraft-Nalton circuit has all capacitors of 0.06 μf. The secondary voltage of the supply transformer is 100 kV at a frequency of 150 Hz. If the load current is 1mA. 8

Determine :

(i) Voltage regulation

(ii) The ripple

(iii) The optimum number of stages for maximum output voltage

(iv) The maximum output voltage

(c) Describe with neat diagrams the principle of operation application and limitations of "Van De Graff" generator. 8

(d) What is cascaded transformer? Explain why cascading is done. Describe with neat diagram of three stage cascaded transformer. 8

Unit-IV

4. (a) Write a basic principle of Electrostatic Voltmeters. 4

(b) What are problems associated with measurement of very high impulse voltages? Explain how these can be taken care of during measurement. 8

(c) Discuss various resistance potential dividers and compare their performance of measurement of impulse voltages. 8

(d) Explain capacitance voltage transformer method for the measurement of high AC voltage? 8

Unit-V

5. (a) Define indirect stroke of lightning. 4

(b) Draw a neat diagram of high voltage Schering Bridge

and analysis it for balanced condition. Assuming series equivalent representation of insulating material. 8

(c) Explain the loss factor $(\tan\delta)$ and capacitance measurement for preventive testing of insulation? 8

(d) What is surge arrester? Explain its function as a shunt protective device. 8